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Unitary matrix

In linear algebra, an invertible complex square matrix U is **unitary** if its matrix inverse U^{-1} equals its conjugate transpose U^* , that is, if

$$U^*U = UU^* = I,$$

where I is the identity matrix.

In physics, especially in quantum mechanics, the conjugate transpose is referred to as the Hermitian adjoint of a matrix and is denoted by a dagger ($\dagger$), so the equation above is written

$$U^\dagger U = UU^\dagger = I.$$

A complex matrix U is **special unitary** if it is unitary and its matrix determinant equals 1.

For real numbers, the analogue of a unitary matrix is an orthogonal matrix. Unitary matrices have significant importance in quantum mechanics because they preserve norms, and thus, probability amplitudes.

Properties

For any unitary matrix U of finite size, the following hold:

- Given two complex vectors $\mathbf{x}$ and $\mathbf{y}$, multiplication by U preserves their inner product; that is, $\langle U\mathbf{x}, U\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$.
- U is normal ($U^*U = UU^*$).
- U is diagonalizable; that is, U is unitarily similar to a diagonal matrix, as a consequence of the spectral theorem. Thus, U has a decomposition of the form $U = VDV^*$, where V is unitary, and D is diagonal and unitary.
- $|\det(U)| = 1$. That is, $\det(U)$ will be on the unit circle of the complex plane.
- Its eigenspaces are orthogonal.
- U can be written as $U = e^{iH}$, where e indicates the matrix exponential, i is the imaginary unit, and H is a Hermitian matrix.

For any nonnegative integer n , the set of all $n \times n$ unitary matrices with matrix multiplication forms a group, called the unitary group $U(n)$.

Every square matrix with unit Euclidean norm is the average of two unitary matrices.^[1]

Equivalent conditions

If U is a square, complex matrix, then the following conditions are equivalent:^[2]

1. U is unitary.
2. U^* is unitary.
3. U is invertible with $U^{-1} = U^*$.
4. The columns of U form an orthonormal basis of $\mathbb{C}^n$ with respect to the usual inner product. In other words, $U^*U = I$.
5. The rows of U form an orthonormal basis of $\mathbb{C}^n$ with respect to the usual inner product. In other words, $UU^* = I$.
6. U is an isometry with respect to the usual norm. That is, $\|Ux\|_2 = \|x\|_2$ for all $x \in \mathbb{C}^n$, where $\|x\|_2 = \sqrt{\sum_{i=1}^n |x_i|^2}$.
7. U is a normal matrix (equivalently, there is an orthonormal basis formed by eigenvectors of U) with eigenvalues lying on the unit circle.

Elementary constructions

2×2 unitary matrix

One general expression of a 2×2 unitary matrix is

$$U = \begin{bmatrix} a & b \\ -e^{i\varphi}b^* & e^{i\varphi}a^* \end{bmatrix}, \quad |a|^2 + |b|^2 = 1,$$

which depends on 4 real parameters (the phase of a , the phase of b , the relative magnitude between a and b , and the angle φ). The form is configured so the determinant of such a matrix is

$$\det(U) = e^{i\varphi}.$$

The sub-group of those elements U with $\det(U) = 1$ is called the special unitary group $SU(2)$.

Among several alternative forms, the matrix U can be written in this form:

$$U = e^{i\varphi/2} \begin{bmatrix} e^{i\alpha} \cos \theta & e^{i\beta} \sin \theta \\ -e^{-i\beta} \sin \theta & e^{-i\alpha} \cos \theta \end{bmatrix},$$

where $e^{i\alpha} \cos \theta = a$ and $e^{i\beta} \sin \theta = b$, above, and the angles $\varphi, \alpha, \beta, \theta$ can take any values.

By introducing $\alpha = \psi + \delta$ and $\beta = \psi - \delta$, has the following factorization:

$$U = e^{i\varphi/2} \begin{bmatrix} e^{i\psi} & 0 \\ 0 & e^{-i\psi} \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} e^{i\delta} & 0 \\ 0 & e^{-i\delta} \end{bmatrix}.$$

This expression highlights the relation between 2×2 unitary matrices and 2×2 orthogonal matrices of angle θ .

Another factorization is^[3]

$$U = \begin{bmatrix} \cos \rho & -\sin \rho \\ \sin \rho & \cos \rho \end{bmatrix} \begin{bmatrix} e^{i\zeta} & 0 \\ 0 & e^{i\zeta} \end{bmatrix} \begin{bmatrix} \cos \sigma & \sin \sigma \\ -\sin \sigma & \cos \sigma \end{bmatrix}.$$

Many other factorizations of a unitary matrix in basic matrices are possible.^{[4][5][6][7][8][9]}

See also

- [Hermitian matrix](#) and [Skew-Hermitian matrix](#)
- [Matrix decomposition](#)
- [Orthogonal group \$O\(n\)\$](#)
- [Special orthogonal group \$SO\(n\)\$](#)
- [Orthogonal matrix](#)
- [Semi-orthogonal matrix](#)
- [Quantum logic gate](#)
- [Special Unitary group \$SU\(n\)\$](#)
- [Symplectic matrix](#)
- [Unitary group \$U\(n\)\$](#)
- [Unitary operator](#)

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External links

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